








Modelling Financial Markets

Chapter 8: Backtesting and Evaluating Risk Forecasts



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How Can We Tell Whether a Risk Forecast Is Useful?

Three groups of questions

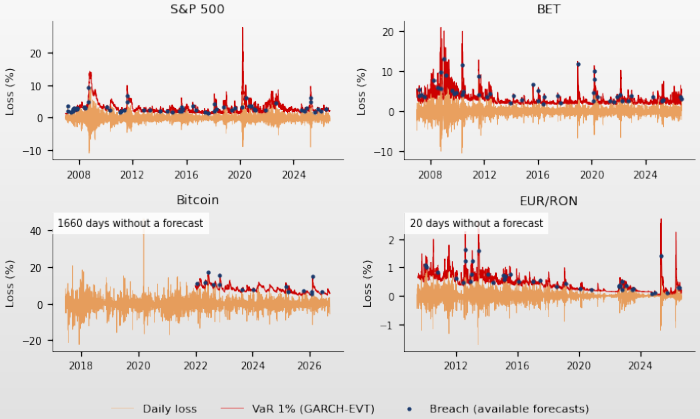
- **Calibration**
  - is a VaR (Value-at-Risk) or ES (Expected Shortfall) forecast right, day after day?
  - can a backtest tell?
- **Comparison**
  - which of several forecasts is better?
  - is the gap significant?
- **Decision**: what can we conclude when the tail holds only a few days a year?

What you should be able to do

- test the number and the timing of VaR breaches and state the size and power of each test at your T
- backtest ES with residuals, Z_2 by simulation and statistics built on the PIT (probability integral transform)
- rank (VaR, ES) forecasts with a consistent score, DM (Diebold–Mariano) tests and the MCS (model confidence set)
- judge what a distribution-free (conformal) VaR does and does not guarantee

Worked exercises: Seminar 8, A1 (Kupiec), A2 (Christoffersen), A4 (Z_2), B2 (FZ0 and DM); readings and derivations: ▸ Appendix

Four Markets, One Risk Model (1/2)



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Four Markets, One Risk Model (2/2)

- Orange: daily losses; red: VaR 1% forecast by GARCH-EVT; blue dots: breaches ($L_t > \mathrm{VaR}_t$)
- The VaR moves with volatility: it doubles or triples in 2008, March 2020 and the 2022 crypto crash
- GARCH-EVT VaR 1% breach rate: 1.15% on the S&P 500 and 0.97% on BET (target: 1%)
- On Bitcoin and EUR/RON, part of the GARCH forecasts are withheld (slide on withheld forecasts)
- EUR/RON has a much smaller scale: a managed exchange rate, long calm spells, rare jumps
- Question for the chapter: is a rate close to 1% enough to call a model good?

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Losses, VaR and ES: the Objects We Test

- **Backtesting**: compare risk forecasts made with the information $\mathcal{F}_{t-1}$ with the outcomes that followed; $\mathcal{F}_{t-1}$: everything known at the close of day $t-1$
- Daily log return $X_t = r_t = \ln(P_t/P_{t-1})$, P_t: closing price on day t; daily **loss** $L_t = -r_t$ (positive number = money lost)
- **VaR**: the loss exceeded with probability α, minus the α-quantile q_α of the return (Chapter 7)
  - $\mathrm{VaR}_\alpha(X) = -\inf\{x : P(X \le x) > \alpha\} = -q_\alpha(X)$; inf: the smallest x at which the distribution function exceeds α
  - equivalently, the $(1-\alpha)$-quantile of the loss L_t; e.g. VaR 1% = 3%: the loss should exceed 3% on about 1 day in 100
- **ES**: the average of the quantiles q_u over all tail levels $u \le \alpha$, i.e. the mean loss beyond VaR
  - $\mathrm{ES}_\alpha(X) = -\frac{1}{\alpha}\int_0^\alpha q_u(X)\,du = E\left[L_t \mid L_t > \mathrm{VaR}_\alpha\right]$ (continuous case)
  - e.g. Student-t losses with $\nu = 4$, standard deviation 1% and $\alpha = 2.5\%$: VaR = 1.96%, ES = 2.82%; always $\mathrm{ES} \ge \mathrm{VaR}$

Breaches and What a Correct VaR Implies

- Levels are tail probabilities: VaR 1% ($\alpha = 0.01$), VaR 2.5% and ES 2.5% ($\alpha = 0.025$); backtests use one-day forecasts
- A **breach** (exception) occurs when the loss exceeds the forecast
  - $I_t = 1\{L_t > \mathrm{VaR}_t\}$, with target rate α; $1\{\cdot\}$: indicator, 1 if the condition holds, 0 otherwise; VaR_t: the forecast made on day $t-1$ for day t
- Ideal **conditional calibration**: whatever we knew yesterday, today's breach probability is exactly α (Christoffersen (1998))
  - $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$ for all t, so the I_t are i.i.d. Bernoulli(α): independent coin flips that equal 1 with probability α
- Two testable consequences
  - **Unconditional coverage**: about αT breaches in T days
  - **Independence**: breaches do not cluster

Six Ways to Forecast Tail Losses

Model	One-day VaR and ES from
HS	historical simulation: the empirical quantile of order $1-\alpha$ of the last 1000 losses
Normal	the Normal distribution with the window mean and standard deviation
Student-t	a Student-t fitted to the window (location, scale, degrees of freedom ν)
GARCH-t	GARCH (generalised autoregressive conditional heteroskedasticity, Chapter 5): $\sigma_t^2 = \omega + a\,\varepsilon_{t-1}^2 + b\,\sigma_{t-1}^2$, with $\varepsilon_{t-1} = r_{t-1} - \mu$; a: reaction to yesterday's shock; b: persistence; Student-t shocks
FHS	filtered HS: GARCH volatility $\times$ empirical quantiles of $z_t = \varepsilon_t/\sigma_t$
GARCH-EVT	EVT (extreme value theory) tail: GARCH volatility $\times$ a GPD (generalised Pareto) tail fitted to the largest 10% of standardised losses (McNeil & Frey (2000))

- **Unconditional** (HS, Normal, Student-t): one distribution for the whole 1000-day window; every day in the window has the same weight, so the forecast reacts slowly to a volatility burst
- **Conditional** or **filtered** (GARCH-t, FHS, GARCH-EVT): a scale σ_t updated daily by GARCH $\times$ a standardised tail

How Are Forecasts Evaluated?

- Four series, each on its own calendar
  - S&P 500 and BET (Bucharest Stock Exchange index): closes
  - Bitcoin (trades every day); EUR/RON: BNR (National Bank of Romania) reference rate
- Rolling window of 1000 observations
  - each forecast uses data up to the previous day only
- Refits every 20 days
  - GARCH, Student-t and EVT parameters; volatility updated daily
- Evaluation days T, until 18.09.2026
  - S&P 500 4 959 and BET 4 937 (from 2007), EUR/RON 4 352 (from 2009), Bitcoin 3 384 (from 2017)

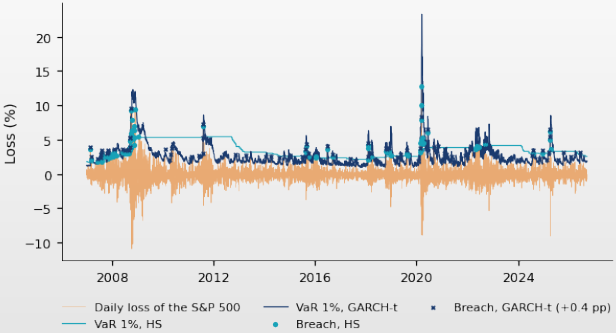
GARCH Forecasts Withheld on Bitcoin and EUR/RON

- GARCH-t, FHS and GARCH-EVT forecasts are reported only on days when the GARCH fit gives a finite VaR
- Bitcoin: 1 660 of 3 384 evaluation days have no such forecast
- EUR/RON: 20 of 4 352 evaluation days have no such forecast
- On these two markets, full-sample results are reported only for HS, Normal and Student-t
- The MCS is not computed on Bitcoin and EUR/RON, because it needs all six models
- Comparisons that include the GARCH models use the S&P 500 and BET, where every forecast exists
- In tables, "n/a" marks a withheld GARCH result; charts mark it with "—"

What a Backtest Can and Cannot Identify

- Each backtest checks one implication of conditional calibration
  - Kupiec: the breach rate; Christoffersen: first-order memory
  - duration tests: the spacing of breaches; DQ (Dynamic Quantile): predictability from chosen regressors
- H_0 is a **joint** hypothesis on the whole forecasting procedure
  - distribution, volatility dynamics, estimated parameters, window length and refit rule
  - A rejection does not say which component failed
- One outcome per day: evidence accumulates only over many days
  - the textbook χ^2 limits treat VaR_t as known
- Non-rejection is not validation: it is informative only if the test has power against the relevant alternatives at the available T

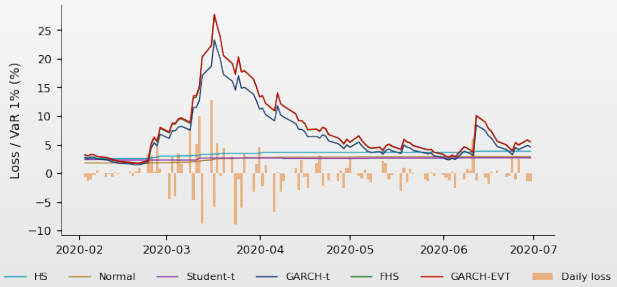
S&P 500: Two VaR Forecasts, 2007–2026



- VaR 1%: HS in teal, GARCH-t in blue; orange: daily losses; markers: breaches of each model
- HS is a step function
  - it jumps only when extreme days enter or leave the 1000-day window
  - GARCH-t rises the day after volatility rises
- Almost the same count: HS 83, GARCH-t 88 breaches in 4 959 days
- Similar counts can hide different timing
  - HS breaches arrive in bursts (2008, 2020)
  - so we must also test **when** breaches occur

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March 2020: The Stress Test Nobody Planned



- S&P 500, February–June 2020: daily losses (bars) and the VaR 1% of the six models
- Largest loss: 12.77% on 16.03.2020
  - a correct VaR 1% expects 1.04 breaches in these 104 days
- Breaches observed
  - unconditional: HS 12, Normal 17, Student-t 17
  - conditional: GARCH-t 3, FHS 2, GARCH-EVT 2
- Current volatility mattered more than the tail shape
  - a fat tail without a moving scale (Student-t) did not help

MFM_ch8_var_forecasts

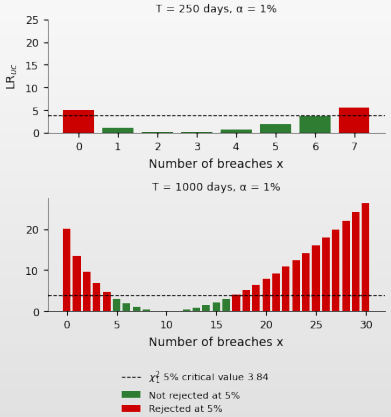
Kupiec (1995): the POF (Proportion of Failures) Test

- Data: $x = \sum_t I_t$ breaches in T days; estimated breach rate $\hat{\pi} = x/T$
- H_0: $\pi = \alpha$ (correct unconditional coverage), against H_1: $\pi \neq \alpha$; π: the true breach probability
- Likelihood ratio statistic (Kupiec (1995)): compares the Binomial likelihood of the observed count under α with that under the observed rate $\hat{\pi}$
  - $\mathrm{LR}_{uc} = -2 \ln \dfrac{(1-\alpha)^{T-x}\alpha^x}{(1-\hat{\pi})^{T-x}\hat{\pi}^x} \xrightarrow{d} \chi^2_1$
  - $\mathrm{LR}_{uc} \geq 0$, equal to 0 when $\hat{\pi} = \alpha$; it grows as $\hat{\pi}$ moves away from α; $\xrightarrow{d} \chi^2_1$: for large T its distribution under H_0 is chi-square with 1 degree of freedom
- Reject at 5% if $\mathrm{LR}_{uc} > 3.84$; two-sided: too few breaches also count as failure; for small T use the exact Binomial(T, α) distribution of x
- Worked example, S&P 500, GARCH-t, 2008: $x = 8$ in $T = 253$ (expected 2.53): $\mathrm{LR}_{uc} = 7.60$, $p = 0.0058$; exact Binomial $p = 2P(X \geq 8) = 0.009$

Which Probability Does This p-Value Sum?

- "Too many breaches?" is a one-sided question: the upper Binomial tail answers it
- Correct coverage is a two-sided question: an overly conservative forecast fails as well
- The worked example's two-sided p-value doubles the upper tail
- For the likelihood-ratio rule, sum the Binomial probabilities of every breach count that triggers rejection: this gives the actual size under H_0 and the power under an alternative
- With discrete breach counts no rule has size exactly 5%: fix the rule before comparing p-values

How Many Breaches Are Acceptable? (1/2)



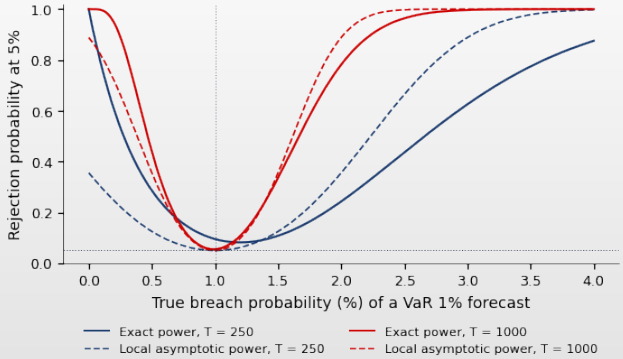
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How Many Breaches Are Acceptable? (2/2)

- LR_{uc} against the number of breaches x, one year (top) and four years (bottom) of VaR 1%
- $T = 250$: the test accepts x from 1 to 6; $T = 1000$: from 5 to 16
- Even $x = 0$ is rejected in one year
  - $\mathrm{LR}_{uc} = -500 \ln 0.99 = 5.03$: a VaR never breached is too conservative
- One year holds about 2.5 informative days for VaR 1%
  - use long samples, two levels ($\alpha = 1\%$ and 2.5%) and several tests

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Power of the POF Test



- Rejection probability at 5% against the true breach rate
  - solid: exact Binomial; dashed: local $\chi^2_1(\lambda)$ approximation, $\lambda = T(\pi - \alpha)^2/(\alpha(1-\alpha))$
- $T = 250$, true rate $\pi = 2\%$
  - local power 36%, exact power 24%
  - a doubled breach rate passes three years in four
- The rule $\mathrm{LR}_{uc} > 3.84$ has actual size 9.5% at $T = 250$
  - $P(x = 0) = 8.1\%$ alone; not an exact 5% test
- 50% power needs
  - $\pi \approx 2.7\%$ at $T = 250$, $\pi \approx 1.7\%$ at $T = 1000$

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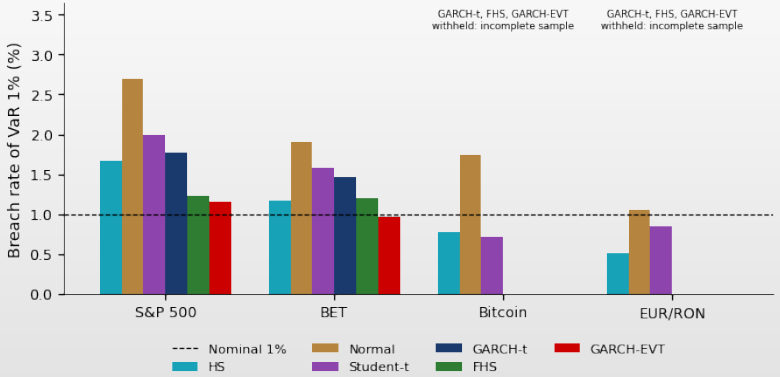
Power of the POF Test: Local Alternatives

- Local alternatives: the true rate approaches α as T grows
  - $\pi_T = \alpha + h/\sqrt{T}$, h: a fixed distance on the $\sqrt{T}$ scale
- Under these alternatives the POF statistic is approximately non-central χ^2
  - $\mathrm{LR}_{uc} = \dfrac{T(\hat{\pi}-\alpha)^2}{\alpha(1-\alpha)} + o_p(1) \xrightarrow{d} \chi^2_1(\lambda), \qquad \lambda = \dfrac{T(\pi-\alpha)^2}{\alpha(1-\alpha)}$
  - $o_p(1)$: a term that vanishes in probability as $T \to \infty$
  - λ: non-centrality; it grows with T and with the squared miss $(\pi-\alpha)^2$
- Reading
  - power $= P(\chi^2_1(\lambda) > 3.84)$: larger λ, more power
  - $T = 250$, $\pi = 2\%$: $\lambda = 2.53$; one year of a doubled rate gives a small λ

Local Power by Hand

- $\chi^2_1(\lambda)$ is the law of $(Z+\sqrt{\lambda})^2$ with $Z \sim N(0,1)$, and $3.84 = 1.96^2$, so
  - power $= P(|Z+\sqrt{\lambda}| > 1.96) = \Phi(\sqrt{\lambda} - 1.96) + \Phi(-\sqrt{\lambda} - 1.96)$, Φ: the $N(0,1)$ distribution function
- $T = 250$, $\pi = 2\%$: $\lambda = 2.525$, $\sqrt{\lambda} = 1.589$
  - power $= \Phi(-0.371) + \Phi(-3.549) = 0.355 + 0.0002 = 0.356$: the 36% of the dashed curve
- The exact power is lower, 24%: the test accepts x from 1 to 6, an event of probability 75.7% when $\pi = 2\%$
  - with 2.5 expected breaches a year, the count is too discrete and too skewed for the Normal approximation behind $\chi^2_1(\lambda)$

Breach Rates Across Models and Markets



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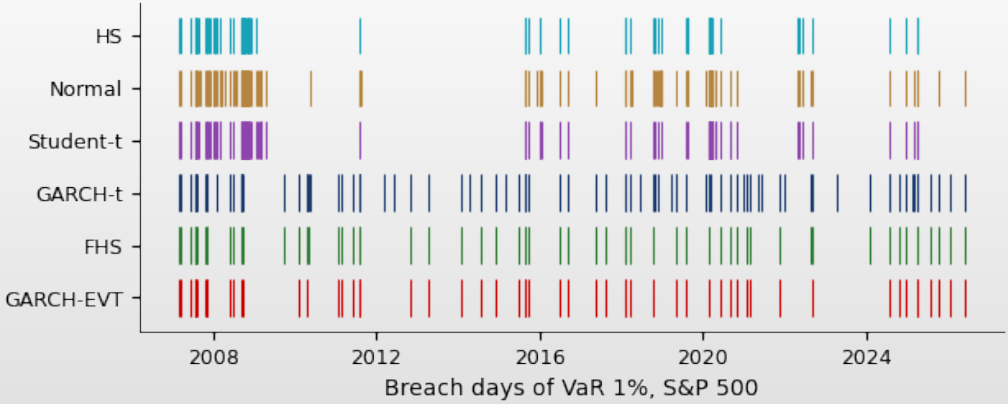
Reading the Breach Rates

- Share of days with $L_t > \mathrm{VaR}_t$ for VaR 1%; the Normal distribution fails except on EUR/RON: S&P 500 2.70%, BET 1.90%, Bitcoin 1.74%
- Same GARCH-t volatility, different tail: GARCH-t 1.77% and 1.46% versus FHS 1.23% and 1.20% (S&P 500, BET): evidence against the symmetric t tail on the loss side
- HS is too conservative on EUR/RON (0.51%): an old crisis in the window keeps its VaR high

Christoffersen (1998): Testing Independence

- Model I_t as a first-order Markov chain with $\pi_{ij} = P(I_t = j \mid I_{t-1} = i)$
- Count transitions n_{ij} (from state i yesterday to state j today); $\hat{\pi}_{01} = \frac{n_{01}}{n_{00}+n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10}+n_{11}}$
- H_0: $\pi_{01} = \pi_{11} = \pi$ (no memory), with $\hat{\pi} = (n_{01} + n_{11}) / \sum n_{ij}$
- Statistic: numerator = likelihood with one breach probability; denominator = likelihood with a probability that depends on yesterday's state
  - $\mathrm{LR}_{ind} = -2 \ln \dfrac{(1-\hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1-\hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi_1^2$
  - Large when $\hat{\pi}_{11}$ (a breach after a breach) differs from $\hat{\pi}_{01}$ (a breach after a calm day); reject at 5% above 3.84
- Conditional coverage: $\mathrm{LR}_{cc} = \mathrm{LR}_{uc} + \mathrm{LR}_{ind} \xrightarrow{d} \chi_2^2$ tests both properties at once (2 degrees of freedom, 5% critical value 5.99)

When Do Breaches Happen? (1/2)



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When Do Breaches Happen? (2/2)

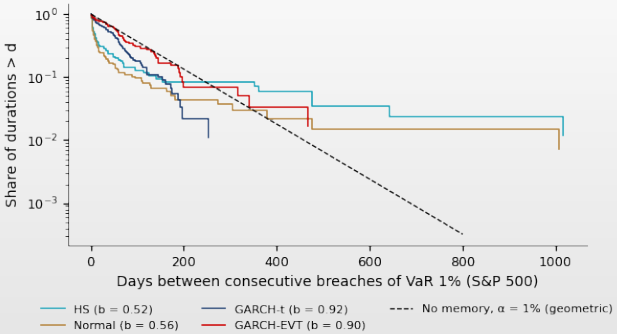
- Each tick: one breach of VaR 1% on the S&P 500; HS, Normal, Student-t: bursts in 2008, 2015–2016, 2018, 2020, 2022; filtered models: no multi-month bursts, yet some back-to-back breaches remain
- A subsample test can miss clustering: HS in 2007–2010, $n_{00} = 928$, $n_{01} = 38$, $n_{10} = 38$, $n_{11} = 3$; $\hat{\pi}_{11} = 0.0732$ versus $\hat{\pi}_{01} = 0.0393$, yet $\mathrm{LR}_{ind} = 0.95$, $p = 0.33$: $\hat{\pi}_{11}$ rests on 3 pairs
- Full sample, p-values of LR_{ind}: HS <0.001, Normal <0.001, Student-t <0.001, GARCH-EVT 0.031, FHS 0.046, GARCH-t 0.095
  - rejected at 5% for every model except GARCH-t; GARCH-EVT: a breach follows a breach with probability 0.053, against 0.011 after a calm day
  - the joint CC test does not reject GARCH-EVT ($p = 0.058$): its correct coverage part dilutes the signal
  - a subsample non-rejection is not a full-sample verdict

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Duration-Based Tests

- Duration D_i: number of days between breach $i-1$ and breach i
- Under a correct VaR, breaches have no memory: D_i is geometric with mean $1/\alpha$ (100 days for $\alpha = 1\%$)
- Christoffersen & Pelletier (2004): approximate the discrete durations by a continuous Weibull model
  - $f(D; a, b) = a^b\, b\, D^{b-1} e^{-(aD)^b}$: the Weibull density of a duration, $a > 0$: rate (inverse scale), $b > 0$: shape; $b = 1$ is the exponential (memoryless) case
  - $b < 1$: short durations are too frequent, i.e. breaches cluster
- H_0: $b = 1$ with a free, a likelihood ratio test (χ^2_1) of memorylessness, not of the mean $1/\alpha$ (coverage is left to Kupiec); first and last durations censored
- Uses the whole spacing pattern: at $T \geq 1\,000$ it detects clustering better than the Christoffersen test; at $T = 250$ it does not (Monte Carlo table)

How Long Between Breaches?



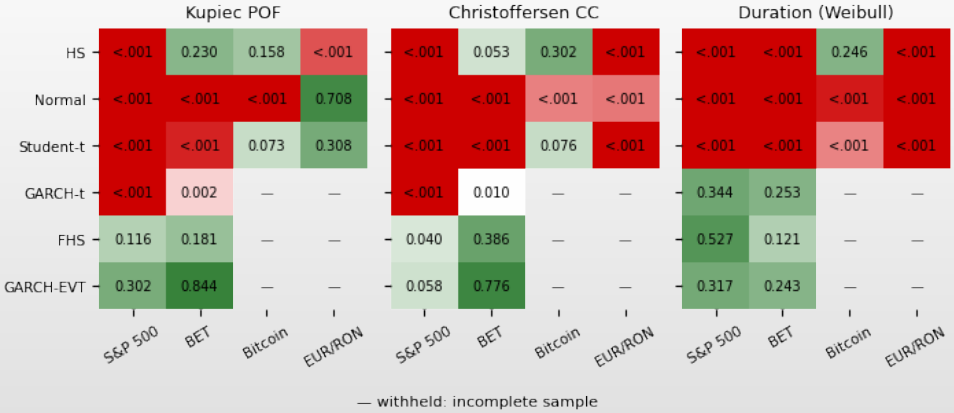
- Share of durations longer than d (log scale); dashed: no memory with $\alpha = 1\%$
- HS: $b = 0.52$, median 7 days, 44% of durations at most 5 days; test $p < 0.001$
- Normal: $b = 0.56$
- GARCH-t: $b = 0.92$, median 45 days, $p = 0.344$
- GARCH-EVT: $b = 0.90$, $p = 0.317$
- The long tail of HS durations: years without breaches after a crisis leaves a high VaR in the window

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Regression-Based Backtests: the DQ Test

- Under correct conditional coverage $\mathrm{Hit}_t = I_t - \alpha$ (the breach indicator, centred) is a martingale difference, i.e. unpredictable from the past: $E[\mathrm{Hit}_t \mid \mathcal{F}_{t-1}] = 0$, $\mathrm{Var}(\mathrm{Hit}_t \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha)$
- So no $\mathcal{F}_{t-1}$-measurable variable may predict it: regress $\mathrm{Hit}_t = X_t\beta + u_t$ with $X_t = (1, \mathrm{Hit}_{t-1}, \ldots, \mathrm{Hit}_{t-4}, \mathrm{VaR}_t)$ and test $\beta = 0$
- Dynamic Quantile statistic (Engle & Manganelli (2004), Wald form with the null variance)
  - $\mathrm{DQ} = \dfrac{\hat{\beta}' X' X \hat{\beta}}{\alpha(1-\alpha)} = \dfrac{\mathrm{Hit}' X (X'X)^{-1} X' \mathrm{Hit}}{\alpha(1-\alpha)} \xrightarrow{d} \chi^2_6$
  - X: the $T \times 6$ matrix of regressors; Hit: the vector of hits; $\hat{\beta}$: OLS (ordinary least squares) estimate; u_t: regression error
  - DQ: explained sum of squares scaled by the null variance; 6 degrees of freedom = 6 regressors; large DQ = predictable hits
- Nests Kupiec and Christoffersen
  - $X_t = 1$ gives the score form $T(\hat{\pi} - \alpha)^2/(\alpha(1-\alpha))$
  - $X_t = (1, \mathrm{Hit}_{t-1})$: a 2-degree-of-freedom counterpart of Christoffersen's conditional coverage test, asymptotically equivalent to LR_{cc} but not numerically equal; testing only the lag coefficient corresponds to LR_{ind}
  - adds power against memory beyond lag 1 and hits that depend on the VaR level

All VaR Tests at a Glance (1/2)



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All VaR Tests at a Glance (2/2)

- The chart: p-values of the Kupiec POF, conditional coverage (CC) and duration (Weibull) tests, VaR 1%; red: rejection, green: no rejection; "—": withheld GARCH forecast
- The DQ test (not in the chart) rejects every S&P 500 model, including FHS ($p < 0.001$) and GARCH-EVT ($p < 0.001$); on BET it does not reject FHS ($p = 0.200$) or GARCH-EVT ($p = 0.279$)
  - hits are predictable from the VaR level and lagged hits, a failure that counting breaches cannot detect
- Different tests examine different failures: read the pattern, not one p-value
  - the unconditional models fail the duration test almost everywhere
  - GARCH-EVT is not rejected by conditional coverage or duration tests on BET

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Size and Power of VaR Backtests: a Monte Carlo

- DGP (data generating process, the simulated truth): GARCH(1,1)-t fitted to the S&P 500 1990–2026 ($a = 0.102$, $b = 0.892$, $\nu = 6.1$); VaR 1%; 1000 paths; rejection rates (%) at 5%

Forecaster	T	POF	Cond. cov.	Duration	DQ
Correct (size)	250	**9.7**	1.1	15.0	8.6
	1000	6.4	3.3	8.0	11.0
	5000	4.9	3.4	4.6	**8.8**
Normal-GARCH (power)	250	**10.6**	4.8	14.8	18.8
	1000	30.2	19.9	8.2	30.2
	5000	83.7	77.2	7.3	70.7
HS, 1000 days (power)	250	51.5	17.0	**12.0**	32.5
	1000	46.4	46.9	68.3	73.8
	5000	57.1	86.6	100.0	99.5

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Size and Power of VaR Backtests: Reading the Table

- The three forecasters
  - Correct: true parameters and true σ_t (rejection rate = size)
  - Normal-GARCH: true σ_t, Normal tail (rejection rate = power)
  - HS, 1000 days: no volatility dynamics (rejection rate = power)
- Conventions
  - an undefined duration p (fewer than 3 durations, 28% of correct samples at $T = 250$) counts as non-rejection
  - Monte Carlo standard error: 0.7 points at 5%, 1.6 at 50%, 0.9 at the observed 9.7% of the bold POF cell
- The bold cells
  - in one year a Normal tail on the right volatility is almost invisible
  - HS clustering is seldom detected by the duration test at $T = 250$
  - DQ rejects a correct model more often than 5% even at $T = 5000$: simulate its critical values

Backtests Under Estimation Risk

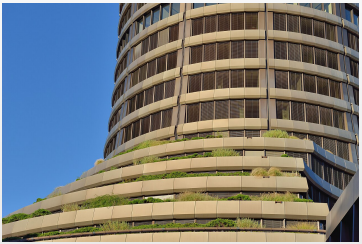
- Forecasts use the parameter estimate $\hat{\theta}$ obtained on R days; the test uses P out-of-sample days; the hits $\hat{I}_t = 1\{L_t > \mathrm{VaR}_t(\hat{\theta})\}$ are not Bernoulli(α) even under a correct model
- The correction depends on the forecasting scheme (fixed, rolling, recursive) and grows with P/R (Escanciano & Olmo (2010); expansion in the Appendix)
- Fixed scheme, 400 GARCH(1,1)-t paths, $R = 1000$: the fit is admissible on 324 paths and not on 76 (simulation results)
- A rejection rate computed only on the 324 admissible paths is conditional on estimation success: Seminar 8, B9 computes it and discusses this selection
- This chapter uses a rolling window ($R = 1000$, refitted every 20 days); a fixed-scheme experiment does not measure the size of these rolling-window tests
- Remedy: reproduce the estimation inside the calibration, e.g. a parametric bootstrap that re-estimates the model on every simulated path (Seminar 8, B9)

The Traffic Light: a Binomial Count Rule

Exceptions	$P(X \leq x)$	Zone
0	8.11%	green
2	54.32%	green
4	89.22%	green
5	95.88%	yellow
7	99.60%	yellow
9	99.97%	yellow
10	99.99%	red

- Banks with internal models backtest daily VaR 1% (Basel Committee on Banking Supervision, BCBS (1996))
  - against actual P&L (profit and loss) over the last 250 days
- $X \sim \text{Binomial}(250, 0.01)$: exceptions of a correct VaR 1% in 250 days
- Yellow starts where $P(X \leq x) \geq 95\%$, red where $P(X \leq x) \geq 99.99\%$
- Only the **number** of exceptions counts, not their size or timing: simple, automatic, hard to game

Backtesting Rules in 1996 and 2019

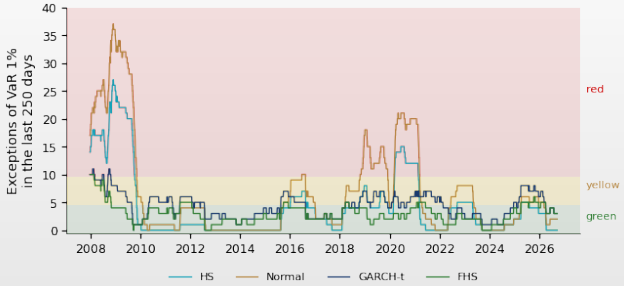


BIS (Bank for International Settlements) tower, Basel: seat of the BCBS secretariat

Photo: Taxiarchos228 (2014); Free Art License; Wikimedia Commons

- BCBS (1996): capital on VaR 1% over 10 days
  - multiplier 3, plus 0.40–0.85 in the yellow zone and plus 1 in the red zone
- BCBS (2019) (FRTB, Fundamental Review of the Trading Book): capital on ES 2.5%, backtesting still on VaR
  - bank-wide: VaR 1%, 250 days; multiplier 1.50 (green, 0–4), 1.70–1.92 (yellow, 5–9), 2.00 (red, 10 or more)
  - trading desk: more than 12 exceptions of VaR 1% or 30 of VaR 2.5% in 12 months move the desk to the standardised approach
  - IMA (internal models approach) approval also requires a P&L attribution test (not covered here)
- VaR counts stay because they are simple
  - ES is not elicitable (section "Elicitability and Scoring Functions")
  - yet ES can be backtested (section "Backtesting Expected Shortfall")

The Traffic Light Through Time



- VaR 1% exceptions in the last 250 days, S&P 500; bands: green, yellow, red zones
- Share of days in red: Normal 22.3%, HS 14.7%, GARCH-t 2.0%, FHS 0.8%
- Peak: HS 27, Normal 37 exceptions
  - in the 250 days ending 01.12.2008; GARCH-t is yellow 46.5% of the time
- Overlapping windows: the penalty lasts a year
  - after a crisis, when capital is already strained

MFM_ch8_traffic_light

FRTB Desk Check on the Last 250 Days (Limits 12 and 30)

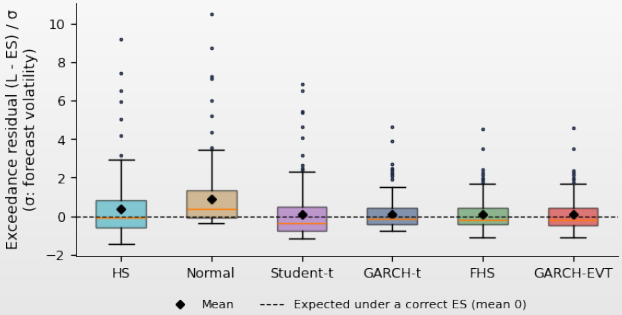
Model	S&P 500	BET	Bitcoin	EUR/RON
HS	0 / 2	4 / 7	3 / 4	1 / 5
Normal	2 / 2	6 / 9	3 / 5	1 / 4
Student-t	0 / 2	4 / 9	1 / 4	5 / 8
GARCH-t	3 / 6	4 / 9	3 / 5	4 / 5
FHS	3 / 4	4 / 6	4 / 5	3 / 4
GARCH-EVT	3 / 3	2 / 7	4 / 5	3 / 4

- Cells: exceptions of VaR 1% / VaR 2.5%
  - last 250 days of each series, to 18.09.2026
  - on these days, GARCH forecasts exist in all four markets
  - every model is well inside the limits
- Illustration only
  - FRTB applies the limits to desk P&L (actual and hypothetical), not to index returns
- The last year was calm
  - a green light today says little about the next crisis

McNeil and Frey (2000): Exceedance Residuals

- ES is an **average** beyond VaR: the information sits on about αT tail days (124 for the S&P 500 at $\alpha = 2.5\%$; 6.25 in one year, so a tail-mean standard error of order $1/\sqrt{\alpha T} \approx 0.40$ standard deviations of the tail losses)
- On each breach day of VaR 2.5%: $e_t = (L_t - \mathrm{ES}_t)/\sigma_t$, with $\sigma_t > 0$ the forecast volatility, known at $t-1$
- If VaR **and** ES are correct: $E[e_t \mid L_t > \mathrm{VaR}_t, \mathcal{F}_{t-1}] = 0$; a wrong VaR changes the conditioning event
- H_0: $E[e_t] = 0$ against H_1: $E[e_t] > 0$ (ES underestimated); one-sided t-statistic
- Few, fat-tailed residuals: p-value from an i.i.d. bootstrap of the centred residuals (McNeil & Frey (2000)), which assumes independent residuals; otherwise a block bootstrap

Exceedance Residuals on the S&P 500



- Distribution of $(L_t - \mathrm{ES}_t)/\sigma_t$ on breach days of VaR 2.5%; diamonds: means
- Means: Normal 0.875, HS 0.363, Student-t 0.082, GARCH-t 0.100, FHS 0.104, GARCH-EVT 0.068
- Bootstrap p
  - Normal $<$0.001, HS $<$0.001, Student-t 0.173, GARCH-t 0.025
  - FHS 0.072, GARCH-EVT 0.157
- Normal: tail mean too low by almost one volatility unit
  - the extreme residuals, up to 10 volatilities, matter most for capital

MFM_ch8_es_backtests

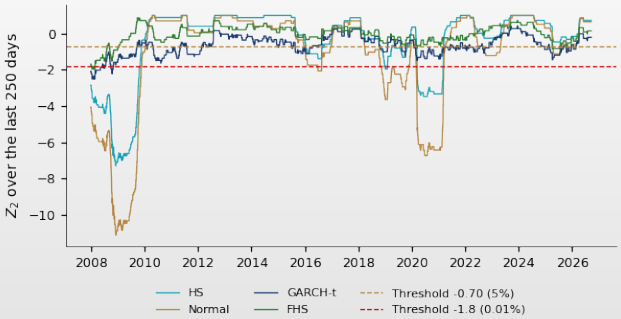
Acerbi and Szekely (2014): Testing ES Directly

- Breach indicator of VaR 2.5%: $I_t = 1\{L_t > \mathrm{VaR}_t^{\alpha}\}$, $N_T = \sum_t I_t$ (the number of breaches in T days), $\alpha = 0.025$; ES_t: the ES forecast for day t
- $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{\mathrm{ES}_t}$: is the average tail loss as large as forecast? (conditional on breaches)
- $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t I_t}{\mathrm{ES}_t}$: tests frequency and size together
- Under H_0 (correct forecast distribution) $E[Z_2] = 0$; underestimated risk gives $Z_2 < 0$ (Acerbi & Szekely (2014))
- For $T = 250$ and many distributions, the 5% and 0.01% critical values are close to -0.70 and -1.8: reference thresholds for an ES traffic light

Getting p-Values by Simulation

- The null distribution of Z_2 depends on the forecast distributions F_t (the distribution function of L_t forecast by the model on day $t-1$)
- Algorithm (repeat $M = 2000$ times)
  - For each day draw $U_t \sim \text{Uniform}(0,1)$ and set $L_t^* = F_t^{-1}(U_t)$, a loss drawn from the forecast (F_t^{-1}: the quantile function)
  - Compute Z_2^* from the simulated losses with the same VaR and ES forecasts
- p-value = share of simulated Z_2^* at or below the observed Z_2
- Only tail draws matter: for HS and FHS we store the 25 largest window losses of each day
- S&P 500, full sample: $Z_2 = -0.664$ for GARCH-t ($p < 0.001$) versus -0.079 for FHS ($p = 0.209$); 5% critical value for $T = 4\,959$ about -0.150, closer to 0 than -0.70 for $T = 250$

Z_2 Over Rolling 250-Day Windows



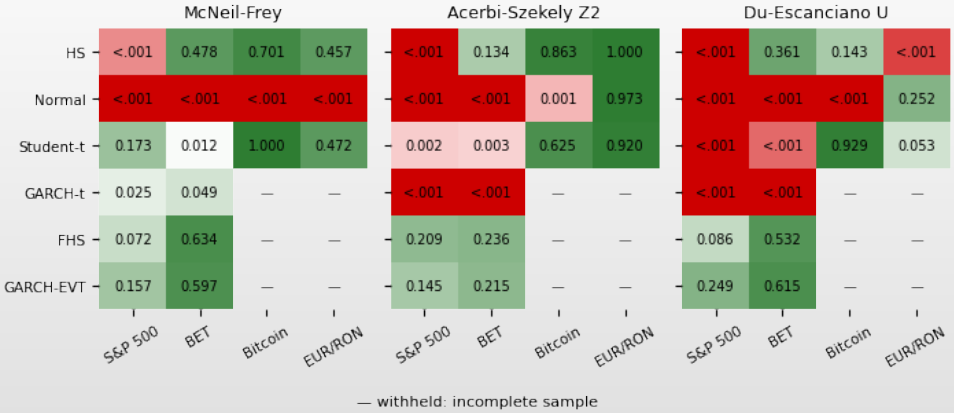
- Z_2 of ES 2.5%, S&P 500, last 250 days
  - reference thresholds -0.70 and -1.8
- Minimum of each curve and the end of its window
  - Normal -11.13 (04.12.2008), HS -7.30 (01.12.2008)
  - GARCH-t -2.52 (05.02.2008), FHS -1.97 (17.01.2008)
- Windows below -1.8
  - Normal 21.9%, HS 15.8%, GARCH-t 3.0%, FHS 0.7%
  - GARCH-t below -0.70 in 44.3% of windows
- Daily checks are repeated tests
  - false alarms accumulate (sequential monitoring, Appendix)

MFM_ch8_es_backtests

Du and Escanciano (2017): Cumulative Violations

- PIT (probability integral transform): $u_t = F_t(L_t)$, the forecast probability of a loss at most L_t; uniform on $(0,1)$ if the forecast distribution F_t is correct
- Cumulative violation: $H_t = \frac{1}{\alpha}\left(u_t - (1-\alpha)\right) 1\{u_t > 1-\alpha\}$: how deep into the tail the loss went
- Under H_0: $E[H_t] = \alpha/2$ and $\mathrm{Var}(H_t) = \alpha(1/3 - \alpha/4)$
- Unconditional test: $U = \sqrt{T}\,(\bar{H} - \alpha/2)/\sqrt{\alpha(1/3-\alpha/4)} \xrightarrow{d} N(0,1)$, with $\bar{H}$ the sample mean of H_t; $U > 1.645$: losses go deeper into the tail than forecast (one-sided, 5%)
  - the heat map on the next slides reports the two-sided $p = 2\Phi(-|U|)$: $U > 0$ means deeper losses than forecast (risk under-predicted), $U < 0$ shallower losses (risk over-predicted)
- Conditional test: Box–Pierce statistic on the first 5 autocorrelations of H_t, χ^2_5 (Du & Escanciano (2017))

All ES Tests at a Glance (1/2)



MFM_ch8_es_backtests

All ES Tests at a Glance (2/2)

- At the 5% significance level, FHS and GARCH-EVT are not rejected by McNeil–Frey, Z_2 and Du–Escanciano U separately on S&P 500 and BET
- Different nulls and inputs, different decisions: Student-t on the S&P 500 has McNeil–Frey $p = 0.173$ but Z_2 $p = 0.002$; these are calibration tests, not rankings
- GARCH-t is rejected by Z_2 on S&P 500 ($p < 0.001$) and BET ($p < 0.001$)
- Read U with its sign: on EUR/RON, HS has $U = -3.68$ ($p < 0.001$), a rejection because losses stay **shallower** than forecast; this matches Z_2 ($p = 1.000$) and its VaR 1% breach rate of 0.51%

MFM_ch8_es_backtests

Multinomial VaR Backtests: an Implicit ES Test

- ES 2.5% averages the quantiles beyond VaR 2.5%: test several VaR levels at once (Kratz, Lok & McNeil (2018))
- $N = 4$ tail levels $\alpha_j = 2.5\%\,(1-(j-1)/4)$: 2.5%, 1.875%, 1.25%, 0.625%; each day falls in one of 5 cells via the PIT u_t
- Pearson statistic $S = \sum_{j=0}^{4}(O_j - Tp_j)^2/(Tp_j) \xrightarrow{d} \chi^2_4$: O_j observed and Tp_j expected days in cell j, cell probabilities $p_0 = 97.5\%$, $p_1 = \cdots = p_4 = 0.625\%$; large S = counts far from expected

p-value	S&P 500	BET	Bitcoin	EUR/RON
HS	<0.001	0.725	0.564	0.004
Normal	<0.001	<0.001	<0.001	<0.001
Student-t	<0.001	0.024	0.015	0.172
GARCH-t	<0.001	<0.001	n/a	n/a
FHS	0.003	0.862	n/a	n/a
GARCH-EVT	0.378	0.853	n/a	n/a

- n/a: withheld GARCH result (slide on withheld forecasts)

Multinomial Cell Counts: S&P 500, GARCH-t

PIT interval	Observed days	Expected days
[0, 0.975]	4761	4 835.02
(0.975, 0.98125]	39	30.99
(0.98125, 0.9875]	46	30.99
(0.9875, 0.99375]	52	30.99
(0.99375, 1]	61	30.99

- The first cell lies outside the upper 2.5% loss tail
- Each of the four tail cells contains more days than expected under calibration

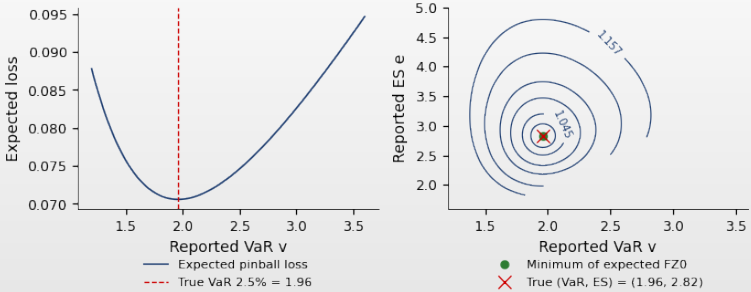
Testing Is Not Ranking: Consistent Scores

- A backtest asks whether a model is **acceptable** (several can pass, all can fail); comparative backtesting asks which is **better** (Nolde & Ziegel (2017))
- Ranking needs a **scoring function** $S(x, L)$, the loss of reporting x when L occurs; compare $\bar{S} = \frac{1}{T}\sum_t S(x_t, L_t)$, lower is better
- S is **consistent** for $\theta(F)$, and θ is **elicitable**, if $\theta(F) = \arg\min_x E_F[S(x, L)]$ (Gneiting (2011))
  - the truth minimises the expected score; ranking needs it to be the **unique** minimiser (strict consistency)
  - $\theta(F)$: the target functional (e.g. VaR) of the loss distribution F; E_F: expectation when $L \sim F$
- VaR: pinball loss $S(v, L) = (1\{L > v\} - \alpha)(L - v)$, v: the VaR forecast
  - if $E|L| < \infty$ and the density $f > 0$ at the quantile:
    $$\frac{\partial}{\partial v} E_F[S(v, L)] = F(v) - (1 - \alpha) = 0 \;\Rightarrow\; v = F^{-1}(1 - \alpha) = \mathrm{VaR}_\alpha, \text{ and } \partial^2/\partial v^2 = f(v) > 0$$
- ES is **not** elicitable on its own: no scoring function of the reported ES forecast e alone is minimised by the true ES (Gneiting (2011))

VaR and ES Together: Fissler and Ziegel (2016)

- The pair (VaR, ES) is jointly elicitable: Fissler & Ziegel (2016) characterise the whole family of consistent scoring functions (the FZ family)
- FZ0 loss, the zero-homogeneous member of the FZ family (Patton, Ziegel & Chen (2019)), with losses positive, $0 < v \leq e$, $E|L| < \infty$ and a true ES above 0
  - $\mathrm{FZ0}(v, e; L) = \dfrac{1\{L > v\}(L - v)}{\alpha\, e} + \dfrac{v}{e} + \ln e - 1$
  - v, e: the VaR and ES forecasts (positive losses); L: the realised loss
- First term: penalty for breaches, scaled by ES; last terms: penalty for reporting a large ES
- Example, one day, $\alpha = 2.5\%$, forecasts $v = 3\%$, $e = 3.9\%$
  - loss $L = 4.5\%$ (a breach): $\mathrm{FZ0} = 15.385 + 1.130 = 16.515$
  - loss $L = 1\%$ (no breach): only $v/e + \ln e - 1 = 0.769 + 1.361 - 1 = 1.130$
- Units: here v, e, L are in percent; only $\ln e$ depends on the unit
  - in decimals ($v = 0.03$, $e = 0.039$, as in the code) the same day scores -3.475 (no breach) and 11.910 (breach): every level moves by $\ln 100 = 4.605$
  - differences between models do not depend on the unit (percent or decimals); this is why the mean FZ0 losses in the code are negative (S&P 500, FHS: -3.551)
- Lower average FZ0 = better joint VaR and ES forecast

Elicitability in Pictures



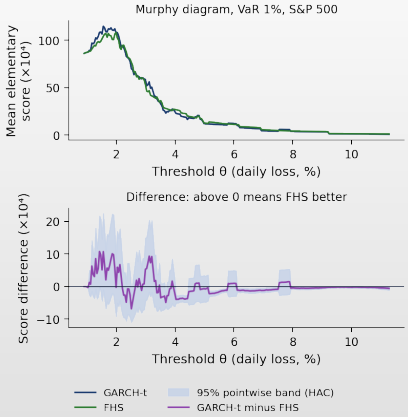
- Student-t losses with $\nu = 4$ and unit variance, $\alpha = 2.5\%$; expected losses computed on 2 million draws
- Left: the expected pinball loss is lowest at the true VaR 2.5%, 1.96; right: the expected FZ0 loss is lowest at (1.96, 2.82), the true (VaR, ES) up to the grid
- A forecaster who knows the truth has no incentive to report anything else: this is what makes ranking by average score meaningful

MFM_ch8_elicitability

Is the Ranking Robust to the Score? Elementary Scores

- For a quantile there are infinitely many consistent scores: $S(v, L) = (1\{L \le v\} - \tau)(g(v) - g(L))$, any increasing function g, $\tau = 1 - \alpha$ (the quantile level of the loss)
- Each is a mixture of **elementary scores** (Ehm, Gneiting, Jordan & Krüger (2016))
  - $S_\theta(v, L) = (1\{L < v\} - \tau)\,(1\{\theta < v\} - 1\{\theta < L\})$, $\theta \in \mathbb{R}$; θ: a threshold on the loss scale; S_θ penalises a forecast only when v and L fall on different sides of θ; the pinball loss is $\int S_\theta \, d\theta$
- Forecast A beats B under **every** consistent score iff $\bar{S}_\theta(A) \le \bar{S}_\theta(B)$ for all θ, with $\bar{S}_\theta$ the average elementary score over the T days
- **Murphy diagram**: plot $\bar{S}_\theta$ against θ; crossing curves mean the ranking depends on the chosen score

Murphy Diagram: GARCH-t versus FHS (1/2)



MFM_ch8_comparative

Murphy Diagram: GARCH-t versus FHS (2/2)

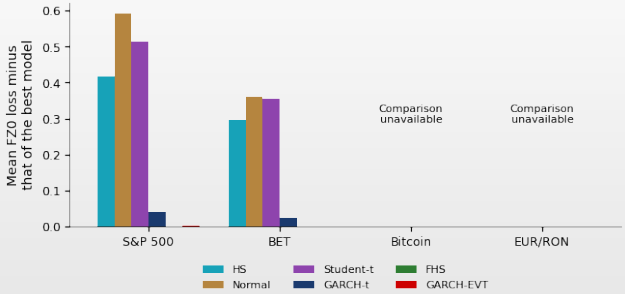
- S&P 500, VaR 1%, 2007–2026; θ from 0.9% to 11.3%; pinball ($\times 10^4$): GARCH-t 3.852, FHS 3.861, a tie under the usual score
- Below $\theta = 5\%$ the curves cross: FHS better on 47% of thresholds, band containing zero on 86%; above 5% GARCH-t lower on 89%: no dominance
- The 95% bands use HAC (heteroskedasticity- and autocorrelation-consistent) standard errors and are pointwise, not a simultaneous test of dominance: state the score before looking at the results

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The Diebold–Mariano Test on FZ0 Losses

- Loss differential of models A and B: $d_t = \mathrm{FZ0}_{A,t} - \mathrm{FZ0}_{B,t}$
- H_0: $E[d_t] = 0$ (equal predictive accuracy), Diebold & Mariano (1995); the statistic is the mean differential $\bar{d}$ divided by its standard error
- $$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{LRV}}(d_t)/T}} \xrightarrow{d} N(0,1)$$
- $\widehat{\mathrm{LRV}}$: long-run variance with Newey–West weights (Newey & West (1987)), 9 lags; tail losses cluster, so d_t is autocorrelated
  - $\widehat{\mathrm{LRV}} = \hat{\gamma}_0 + 2\sum_{k=1}^{q}\left(1 - \frac{k}{q+1}\right)\hat{\gamma}_k$, $\hat{\gamma}_k$: autocovariance of d_t at lag k; lag rule $q = \lfloor 4(T/100)^{2/9} \rfloor = 9$ for $T = 4\,959$
- $\mathrm{DM} > 0$: model A has the larger average loss (worse); $|\mathrm{DM}| > 1.96$: significant at 5%

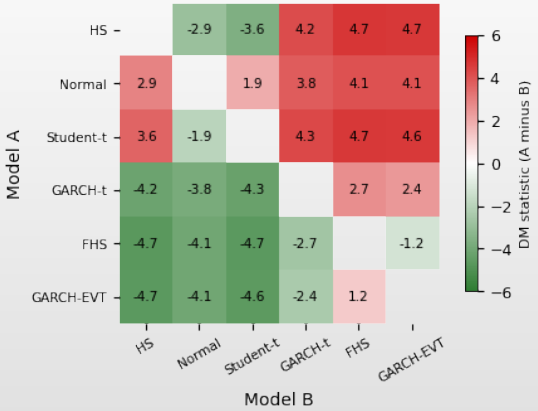
Average FZ0 Loss by Model



- Mean FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts minus that of the best model in each market
- Lowest FZ0 loss: FHS on S&P 500 and BET
- Comparison on the complete evaluation sample
  - On S&P 500 and BET, compare all six models on the same evaluation dates
- The mean FZ0 loss of GARCH-t exceeds that of FHS by 0.040 on the S&P 500
  - the size of a gap does not tell whether the data separate the models: we test it

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Pairwise Diebold–Mariano Tests, S&P 500 (1/2)



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Pairwise Diebold–Mariano Tests, S&P 500 (2/2)

- DM of model A (row) minus B (column) on FZ0 losses; red: A worse, green: A better
- Against FHS
  - HS $\mathrm{DM} = 4.69$, GARCH-t $\mathrm{DM} = 2.66$: both significantly worse
  - GARCH-EVT $\mathrm{DM} = 1.24$: indistinguishable
- Normal minus HS: $\mathrm{DM} = 2.88$: a slow empirical quantile still beats a thin tail
- Fifteen pairwise tests: a multiple-testing problem
  - which models survive **jointly**? (model confidence set, MCS)

MFM_ch8_comparative

Comparing Forecasting Methods: Giacomini–White

- DM targets the population accuracy of **models**
  - with rolling 1000-day windows and refits we compare **forecasting methods**; estimation error never vanishes
- Giacomini & White (2006): H_0: $E[d_t \mid \mathcal{F}_{t-1}] = 0$ for the score differential $d_t = \mathrm{FZ0}_{A,t} - \mathrm{FZ0}_{B,t}$; valid for nested models and finite windows
- Test function $h_{t-1} = (1, d_{t-1})'$: information known at $t-1$ that might predict which method wins; $Z_t = h_{t-1} d_t$
  - $\mathrm{GW} = T\,\bar{Z}'\hat{\Omega}^{-1}\bar{Z} \xrightarrow{d} \chi^2_2$, $\hat{\Omega} = T^{-1}\sum_t Z_t Z_t'$ (one step ahead); $\bar{Z}$: the mean of Z_t, $\hat{\Omega}$: its estimated covariance; 2 degrees of freedom = 2 instruments; $h_{t-1} = 1$ gives the DM form
- S&P 500 against FHS: GARCH-EVT $\mathrm{GW} = 6.71$ ($p = 0.035$), GARCH-t 7.60 ($p = 0.022$), HS 53.16 ($p < 0.001$)
- GARCH-EVT, a tie under DM, is rejected conditionally
  - equal conditional predictive ability fails for these instruments
  - what drives the rejection needs a separate test

Three-Zone Comparative Backtesting (Nolde and Ziegel)

- Internal model A against a standard model S (here HS), mean FZ0 loss; two one-sided tests at 5% (Nolde & Ziegel (2017))
- H_0^-: A at least as good as S; H_0^+: A at most as good. **Green**: reject H_0^+; **red**: reject H_0^-; **yellow**: neither
- One-sided cut-offs: **green** if $\mathrm{DM} < -1.645$, **red** if $\mathrm{DM} > 1.645$, **yellow** otherwise
  - for example, $\mathrm{DM} = -1.8$ gives the green zone although $|\mathrm{DM}| < 1.96$
- The burden of proof is reversed: the bank must show that its model is better, not the supervisor that it is worse

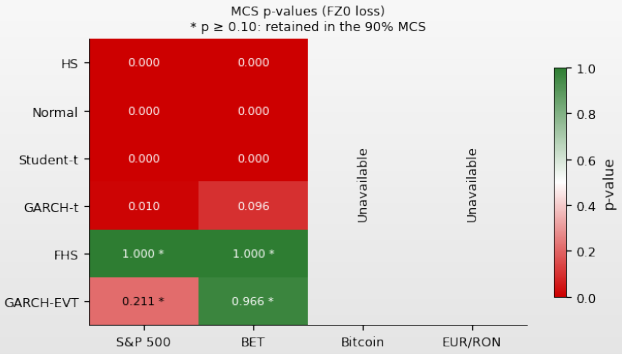
DM (A minus HS)	S&P 500	BET	Bitcoin	EUR/RON
GARCH-t	−4.21	−5.25	n/a	n/a
FHS	−4.69	−5.53	n/a	n/a
GARCH-EVT	−4.67	−5.57	n/a	n/a

- Colour indicates the statistical zone
- n/a: withheld GARCH result (slide on withheld forecasts)

The Model Confidence Set

- Model confidence set, MCS (Hansen, Lunde & Nason (2011))
  - contains the best models with asymptotic probability at least $1 - \alpha_{MCS}$
  - $\alpha_{MCS} = 10\%$ below; under the method's regularity conditions
- Algorithm
  - Test H_0: all models in the current set have equal expected loss
  - If rejected, eliminate the model with the largest standardised average loss and repeat
  - Critical values from a block bootstrap (blocks of 10 days, 1000 replications)
- MCS p-value of a model: the largest test p-value up to its elimination; keep models with $p \geq 0.10$
- A large MCS is an honest answer: the data do not separate the models

Which Models Survive?



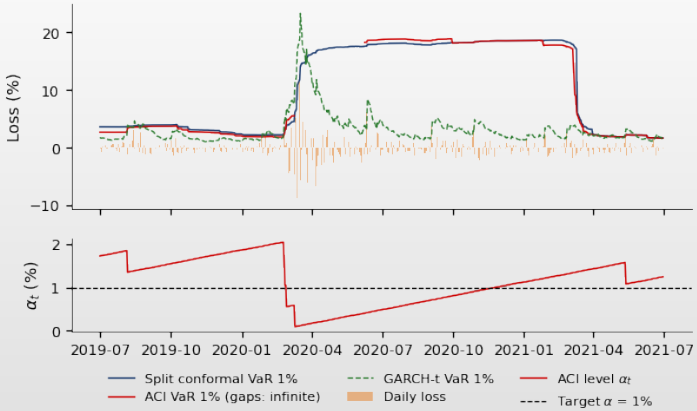
- MCS p-values, FZ0 loss of (VaR 2.5%, ES 2.5%)
  - models with $p \geq 0.10$ form the 90% MCS
- S&P 500: FHS and GARCH-EVT
  - GARCH-EVT MCS p 0.211; GARCH-t out (p 0.010)
- BET: FHS and GARCH-EVT
  - GARCH-t just out (p 0.096)
- Bitcoin and EUR/RON: no MCS, because the GARCH forecasts are withheld (slide on withheld forecasts)

MFM_ch8_comparative

Split Conformal Quantiles: a Distribution-Free Promise

- **Conformal prediction** (Lei et al. (2018), Angelopoulos & Bates (2021))
  - the models so far assume a distribution (Normal, t, GPD) or a stable history (HS)
  - conformal prediction turns any forecast into a quantile with a coverage guarantee
- Score $s_t = L_t/\hat{\sigma}_t$, $\hat{\sigma}_t$ the standard deviation of the previous 250 returns; calibration set: the last $n = 250$ scores, sorted $s_{(1)} \leq \cdots \leq s_{(n)}$
- Conformal VaR: $\mathrm{VaR}_t = \hat{\sigma}_t\, s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$ ($\lceil x \rceil$: smallest integer $\geq x$): today's volatility times the k-th smallest past score; $n = 250$, $\alpha = 0.01$: $k = 249$, the second largest score
- Guarantee if the $n+1$ scores are **exchangeable** (joint law invariant to reordering): $1 - \alpha \leq P(L_t \leq \mathrm{VaR}_t) \leq 1 - \alpha + \frac{1}{n+1}$ (no ties)
- Here the guarantee is only approximate (extensions: ▸ Appendix)
  - volatility clustering breaks exchangeability; $\hat{\sigma}_t$ comes from overlapping rolling windows
  - genuine split conformal fits the score once, on a separate sample

Conformal VaR Through COVID-19 (1/2)



MFM_ch8_conformal

Conformal VaR Through COVID-19 (2/2)

- S&P 500, July 2019–June 2021, VaR 1%
  - top: split conformal, ACI and GARCH-t; bottom: ACI level α_t
  - ACI (adaptive conformal inference) adjusts the level α_t daily after breaches (next slide)
- 20.02–30.04.2020: 50 days, 0.5 breaches expected
  - split conformal 6, ACI 4 (8% of days)
- ACI set VaR $= \infty$ on 37 of these days
  - no further breaches, but an infinite VaR carries no information
- Split conformal VaR 1% stays near 18% for a year
  - two factors stay inflated for about a year: today's $\hat{\sigma}_t$ (a 250-day standard deviation that still contains March 2020) and $s_{(\mathbf{249})}$, the second largest of the last 250 scores, which stays extreme while the crash days are in the calibration set
  - illustrative numbers: if $\hat{\sigma}_t$ doubles from 1.2% to 2.4% and $s_{(\mathbf{249})} = 7$, the VaR moves from 8.4% to 16.8%
  - GARCH-t adapts within days

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Adaptive Conformal Inference (ACI)

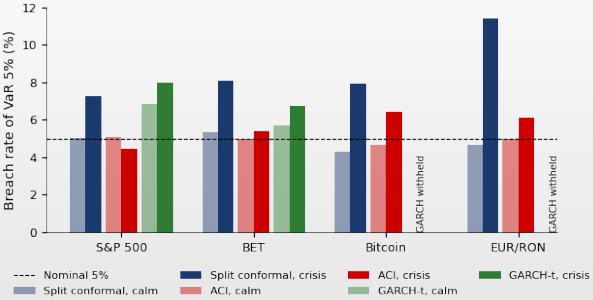


Emmanuel Candès, co-author of adaptive conformal inference

Photo: Cmichel67 (2026); CC BY-SA 4.0; Wikimedia Commons

- Gibbs & Candès (2021): the working level is updated after each day
  - $\alpha_{t+\mathbf{1}} = \alpha_t + \gamma\,(\alpha - \mathrm{err}_t)$, $\mathrm{err}_t = 1\{L_t > \mathrm{VaR}_t\}$
  - α_t: level of day t's conformal quantile; $\gamma > 0$: step size; $\alpha_{\mathbf{1}}$: starting level
- After a breach α_t falls (VaR rises); after calm days it slowly rises again; we use $\gamma = 0.005$
- Guarantee for any sequence: $\left|\frac{1}{T}\sum_t \mathrm{err}_t - \alpha\right| \le \frac{\max(\alpha_{\mathbf{1}}, 1-\alpha_{\mathbf{1}}) + \gamma}{\gamma T}$, a long-run **average**, not each day or each crisis
- If $\alpha_t < 1/(n+1)$, the required quantile does not exist: the VaR becomes infinite

Coverage in Calm and Crisis Periods



- VaR 5%; crisis day: known in advance
  - previous 20-day volatility above the 90th percentile of the previous 1000 days
- Split conformal breach rates
  - all days (S&P 500, BET, Bitcoin, EUR/RON): 5.34%, 5.63%, 4.58%, 5.06%
  - crisis days: 7.29%, 8.08%, 7.95%, 11.43%
- ACI: equity crisis rates near 5%
  - S&P 500 4.49%, BET 5.38%
  - for VaR 1% only by an infinite VaR on 13.4% of all 4 959 S&P 500 evaluation days
  - marginal coverage is not crisis coverage

MFM_ch8_conformal

Case Study (1/8): Can ES Be Backtested Using Only ES Forecasts?

- Bayer & Dimitriadis (2022): can ES forecasts be backtested with nothing but the realised returns and the ES forecasts themselves?
- Why it matters: FRTB puts capital on ES 2.5% (section "The Regulator's View")
  - the ES tests of the section "Backtesting Expected Shortfall" also need VaR forecasts, the volatility or the whole forecast distribution
  - supervisors do not necessarily receive these inputs
  - a rejection of such a joint test does not say which input is wrong
- Idea: a Mincer–Zarnowitz regression of the returns on the ES forecasts, H_0: intercept 0, slope 1 (a correct forecast needs no correction)
  - ES alone is not elicitable (section "Elicitability and Scoring Functions"), but (VaR, ES) is jointly elicitable (Fissler & Ziegel (2016)), so a joint quantile–ES regression can be estimated (Patton, Ziegel & Chen (2019))

Case Study (2/8): the ES Regression

- Returns convention: $\hat{e}_t = -\widehat{\mathrm{ES}}_t < 0$, the ES forecast on the return scale, with $\tau = \alpha = 2.5\%$
  - ES equation: $r_t = \gamma_{\mathbf{1}} + \gamma_{\mathbf{2}}\hat{e}_t + u_t^e$, $\mathrm{ES}_\tau(u_t^e \mid \mathcal{F}_{t-\mathbf{1}}) = 0$
  - $\gamma_{\mathbf{1}}$: intercept, $\gamma_{\mathbf{2}}$: slope; u_t^e: error whose conditional ES is zero, so $\gamma_{\mathbf{1}} + \gamma_{\mathbf{2}}\hat{e}_t$ is the true conditional ES; correct forecasts: $\gamma_{\mathbf{1}} = 0$, $\gamma_{\mathbf{2}} = 1$
- ESR (expected shortfall regression) also needs a quantile equation, because ES is estimated jointly with VaR
  - **Auxiliary ESR**: quantile equation on $(1, \hat{v}_t)$, needs VaR forecasts $\hat{v}_t$; **Strict ESR**: quantile equation on $(1, \hat{e}_t)$, ES forecasts only
  - **Intercept ESR**: $r_t - \hat{e}_t = \gamma_{\mathbf{1}} + u_t^e$, slope fixed at 1; one-sided t-test of $H_{\mathbf{0}}$: $\gamma_{\mathbf{1}} \geq 0$ against $\gamma_{\mathbf{1}} < 0$ (ES underestimated)

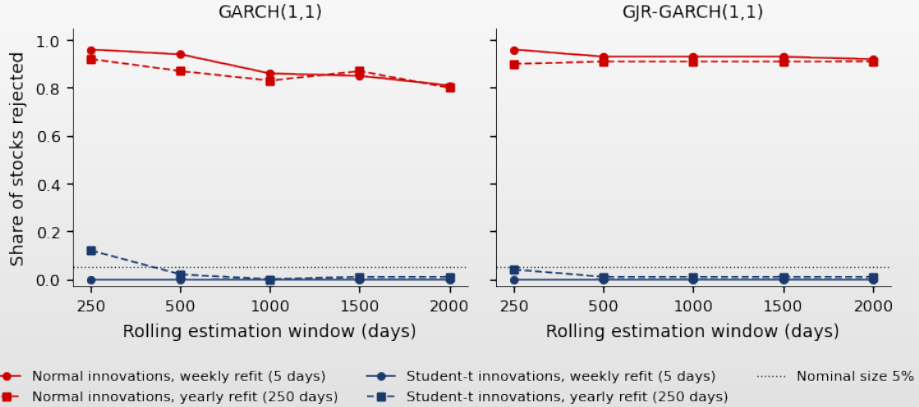
Case Study (3/8): What Is Estimated, Under Which Assumptions

- M-estimator minimising an FZ0-type loss
  - M-estimator: the parameter value that minimises the sample average of a loss; OLS is the squared-error case
  - asymptotics under misspecification (paper, Theorem 1.4)
  - R packages esback and esreg
- Two assumptions to keep in mind
  - Strict and Intercept ESR: the quantile equation is correct only if VaR is an affine function of ES (e.g. pure scale models)
  - The misspecification-robust sandwich covariance assumes a location-scale model for the conditional distribution
  - sandwich covariance: $A^{-1}BA^{-1}$, A: curvature of the average loss, B: variance of its gradient; location-scale model: $r_t = \mu_t + \sigma_t z_t$ with i.i.d. standardised shocks z_t

Case Study (4/8): Size and Power in the Paper's Simulations

- Monte Carlo design (paper, Sec. 2)
  - 10 000 replications, T from 250 to 5 000
  - EGARCH-t (exponential GARCH, Student-t shocks) calibrated to the S&P 500; AR(1)-GARCH(1,1): first-order autoregressive mean, GARCH variance; two Student-t GAS (generalised autoregressive score) models
  - competitors: McNeil & Frey (2000) and the calibration tests of Nolde & Ziegel (2017)
- Size at 5%, EGARCH-t, $T = 250$ and $T = 1\,000$ (Table 1)
  - Strict ESR: 0.09 and 0.05 with the robust covariance, 0.24 and 0.11 with the classical one
  - simple conditional calibration test: 0.29 and 0.14
- Power (Fig. 3): ESR detects what residual tests miss
  - exceedance-residual tests barely react to wrong volatility dynamics or a wrong level $\tilde{\tau} \neq \tau$
  - a bank could report ES for a milder level and pass the older tests

Case Study (5/8): Tails Matter More Than Refits



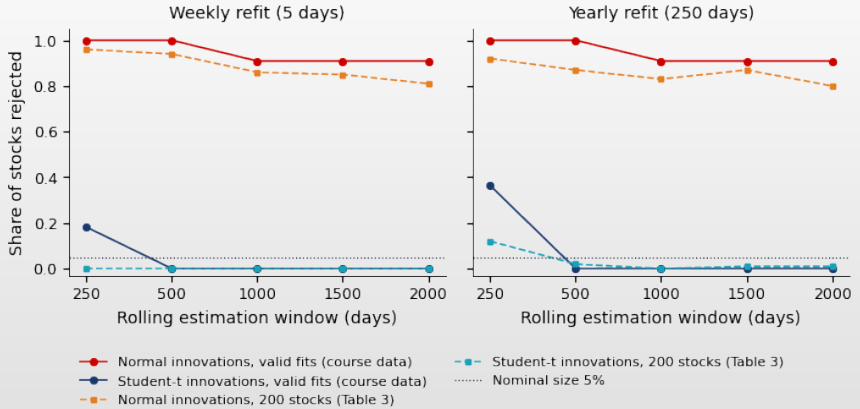
MFM_ch8_esr_case

Case Study (6/8): Tails Matter More Than Refits

- Bayer & Dimitriadis (2022), Table 3: 200 largest S&P 500 stocks, 2010–2019
  - share with ES 2.5% rejected by the one-sided Intercept ESR at 5%
  - windows of 250–2 000 days; refits every 5 (solid) or 250 days (dashed)
- Normal innovations: 0.80–0.96 rejected
  - in all 50 cells of Table 3: 5 windows (250, 500, 1 000, 1 500, 2 000 days) $\times$ 5 refit frequencies (5, 21, 62, 125, 250 days) $\times$ GARCH/GJR-GARCH
  - the chart shows only the 5- and 250-day refits
  - GJR-GARCH (Glosten–Jagannathan–Runkle): GARCH with an extra reaction to negative shocks
- Student-t: at most 0.12
  - up to 0.12 with 250-day windows, at most 0.01 from 1 000 days
- Fat tails and a long window matter most
  - the refit frequency matters only with short windows

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Case Study (7/8): Table 3 Replicated on the Course Data



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Case Study (8/8): Table 3 Replicated on the Course Data

- Solid: 11 large S&P 500 stocks with prices from 2002
  - January 2010 – August 2019 (2 432 days); dashed: Bayer & Dimitriadis (2022), Table 3
  - GARCH(1,1) with a constant mean
  - 25 window–refit combinations (the same 5×5 grid, GARCH only); Intercept ESR with esback
  - each rejected stock adds $1/11 \approx 0.09$ to a share
- Normal: 0.82–1.00 rejected (paper 0.80–0.96)
  - Student-t: no stock rejected with windows of at least 1 000 days (paper: at most 0.01)
  - but 0.09–0.55 at 250 days (paper 0.00–0.12)
- The central result replicates; with 250-day windows the shares reach 0.55, against at most 0.12 in the paper

MFM_ch8_esr_case

An Open Question: Crisis-Conditional Coverage

- Distribution-free VaR guarantees are **marginal**: split conformal needs exchangeability, ACI controls only the long-run average (section "Conformal Prediction for Tail Risk")
- Formal question, with a crisis flag $C_t \in \mathcal{F}_{t-1}$ known ex ante ($C_t = 1$ on a crisis day, 0 otherwise): is the breach rate on crisis days still α?
  - H_0: $P(L_t > \mathrm{VaR}_t \mid C_t = 1) = \alpha$ against H_1: $P(L_t > \mathrm{VaR}_t \mid C_t = 1) \neq \alpha$
  - Falsified by a crisis breach rate outside the Binomial acceptance region, after correcting for dependent breaches and for multiplicity
- One pass of the discovery loop with an AI (artificial intelligence) assistant: an LLM (large language model) such as Claude, ChatGPT or Gemini
  - literature with DOIs checked on Crossref; replicate a known number first
  - three rival explanations: stale scale, changing tail shape, regime-selection artefact
  - pre-register the tests; ask the assistant for a hostile referee report (the prompts: Appendix)
- Stakes: capital is needed exactly on the days when marginal guarantees say least

Mini-Case: Does a GARCH Scale Repair Crisis Coverage?

- Alternative score: the standardised loss $s_t = (L_t + \mu_t)/\sigma_t$ as score
  - μ_t, σ_t: GARCH-t forecasts of the mean and volatility
  - H_0: crisis breach rate of VaR 5% = 5%; exact two-sided Binomial test, Holm correction (a step-down adjustment of the p-values for multiple tests) across the four markets for split conformal and across S&P 500 and BET for the GARCH scale (n/a: withheld GARCH forecasts)

Market	Crisis days	Split conformal	Holm p	GARCH-scaled	Holm p
S&P 500	613	7.01%	0.052	2.77%	0.018
BET	506	8.10%	0.009	3.95%	0.309
Bitcoin	150	8.00%	0.092	n/a	n/a
EUR/RON	245	11.43%	<0.001	n/a	n/a

ai_discovery_case.py

Mini-Case: Reading the Result

- Sample change first
  - the GARCH-scaled scores need 250 calibration days
  - common S&P 500 sample: 4 709 days, 613 crisis days (full sample of the conformal section: 713)
  - split conformal crisis rate: 7.01% (full sample: 7.29%)
- GARCH-scaled crisis rates are below 5% on S&P 500 and BET
  - GARCH scale, Holm across S&P 500 and BET: the 2.77% rate is significantly below 5% on the S&P 500 (p 0.018), the 3.95% on BET is not (p 0.309)
  - Split conformal: Holm rejects on BET (p 0.009) and EUR/RON (p <0.001) at 5%
- Non-rejection does not prove crisis coverage
  - Binomial p-values ignore clustered breaches
  - the crisis samples are small

Which Model Would You Provisionally Select?



National Bank of Romania, Bucharest

Photo: Britchi Mirela (2018); CC BY-SA 4.0; Wikimedia Commons

- **Calibration**
  - FHS and GARCH-EVT are not rejected by Z_2 at 5% and remain in the 90% MCS on S&P 500 and BET (scorecard: Appendix)
- **Comparison**
  - lowest mean FZ0: FHS on S&P 500; FHS and GARCH-EVT are equal to three decimals on BET
- **Unresolved**
  - DQ rejects every S&P 500 model; crisis-conditional coverage fails; one year of data has little power
- **Provisional choice**
  - FHS as the main model, GARCH-EVT as the challenger, both re-examined as evidence accrues
  - a Romanian portfolio (BET and EUR/RON): Seminar 8, C2

Check Your Understanding, then Chapter 9

Retrieval questions

- ⊡ Why does the rule "$\mathrm{LR}_{uc} > 3.84$" reject a correct VaR 1% in about one year in ten?
- ⊡ Why can a model with the right number of breaches still fail?
- ⊡ Why can ES be backtested but not ranked on its own?
- ⊡ What does ACI guarantee?
- ⊡ What does ACI not guarantee?

Chapter 9: realised volatility

- ⊡ Measure daily variance from intraday returns
- ⊡ Evaluate volatility forecasts with robust scores such as QLIKE (quasi-likelihood loss) and the MCS

Numeric Self-Checks

1. Kupiec: $x = 3$ breaches of VaR 1% in $T = 500$ days; compute $\hat{\pi}$ and LR_{uc}, then decide at 5%
2. FZ0: one day, $\alpha = 5\%$, $v = 2\%$, $e = 2.5\%$; compute the score for $L = 3\%$ and for $L = 1\%$, in percent and in decimals
3. Local power: $T = 1000$, $\alpha = 1\%$, true $\pi = 2\%$; compute λ and the power, then compare with the dashed curve of the power chart
4. Z_1 and Z_2: $T = 250$, $\alpha = 2.5\%$, five breaches with $L_t/\mathrm{ES}_t = 1.1,\ 0.9,\ 1.3,\ 1.0,\ 1.2$; compute both and explain their signs

Answers with intermediate steps: Appendix. Worked exercises on the course data: Seminar 8, A1, A4, B2

Further Reading and Resources

- Key papers
  - Fissler & Ziegel (2016): VaR and ES are jointly elicitable, the basis of the FZ0 loss
  - Nolde & Ziegel (2017): comparative backtests and the traffic light for VaR and ES
  - Acerbi & Szekely (2014): model-free ES backtests, including Z_2
- Entropy-based VaR and ES forecasts: Pele, Lazar & Dufour (2017), Pele & Mazurencu-Marinescu-Pele (2019), Pele, Lazar & Mazurencu-Marinescu-Pele (2019)
- When two ES forecasts can be told apart: Pele & Mazurencu-Marinescu-Pele (2026)
  - about one in five pairwise ES comparisons in global markets is precision-fragile
  - a consequence of the small effective tail sample αT
- Textbooks: Franke, Härdle & Hafner (2019) (VaR and backtesting chapters) and Härdle, Chen & Overbeck (2017)
- Quantinar: Machine learning in Financial Risk
- Code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_08; S&P 500, BET, Bitcoin: daily closes from the course data; EUR/RON: BNR reference rate

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Estimation Risk, the Fixed-Scheme Expansion

- Fixed scheme (Escanciano & Olmo (2010)): $\frac{1}{\sqrt{P}} \sum_t (\hat{I}_t - \alpha) = \frac{1}{\sqrt{P}} \sum_t (I_t - \alpha) + \sqrt{P/R}\, D' \sqrt{R}(\hat{\theta} - \theta_0) + o_p(1)$
  - First term: the centred hits with the true parameters θ_0; second term: the effect of the estimation error $\hat{\theta} - \theta_0$
  - $D = E\left[\partial P(L_t > \mathrm{VaR}_t(\theta) \mid \mathcal{F}_{t-1})/\partial\theta\right]$ at θ_0, the sensitivity of the breach probability to the parameters; the second term vanishes only if $P/R \to 0$
  - The estimation term adds a variance of order P/R to $\sqrt{P}(\hat{\pi} - \alpha)$, so χ^2 critical values are too small when P/R is large
- Rolling and recursive schemes change the form of the correction; the expansion assumes smooth $\mathrm{VaR}_t(\theta)$, a $\sqrt{R}$-consistent $\hat{\theta}$ and stationarity
- Remedies: the corrected variance or subsampling of Escanciano & Olmo (2010), or a parametric bootstrap that re-estimates the model on every simulated path

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Estimation Risk: Simulation Results

- Fixed scheme: 400 paths, $R = 1000$, Kupiec test at 5%
- 324 admissible fits; 75 numerically unidentified; 1 fails the finite-variance admissibility condition
- We report only the true-parameter rates, on all 400 paths: a rate on the 324 admissible paths would be conditional on estimation success
- True parameters: 10.0% at $P = 250$, 5.2% at $P = 1000$, 4.0% at $P = 5000$
- Monte Carlo standard error at a 5% rejection probability: about 1.1 percentage points

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Osband's Principle and Identification Functions

- Osband's principle (Fissler & Ziegel (2016)): consistent scores are built from an identification function V, with $E_F[V(x, L)] = 0$ only at the true value
- VaR: $V(v, L) = 1\{L \leq v\} - (1 - \alpha)$; the pinball loss is its integral in v
- (VaR, ES): $V_1 = \alpha - 1\{L > v\}$ and $V_2 = e - v - \frac{1}{\alpha} 1\{L > v\}(L - v)$; their expectations vanish only at the true pair
- The level sets of ES are not convex under mixtures, so no identification function of e alone exists (Seminar 8, A8)
- FZ0 is the member of the FZ family whose differences are invariant to the scale of the losses (Patton, Ziegel & Chen (2019))

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Sequential Monitoring of Risk Forecasts

- A regulator checks the forecasts every day: repeating a fixed-sample test (e.g. rolling Z_2) inflates the probability of at least one false alarm
- Hoga & Demetrescu (2023): monitoring statistics for VaR and ES with boundary functions that keep the probability of any false alarm over the whole horizon below the nominal level
- Compared with the traffic light: evidence accumulates from the start of monitoring instead of a single 250-day snapshot
- The ES regression backtest (Bayer & Dimitriadis (2022)) of the case study can be monitored the same way; both address the one-shot design of the traffic light

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Conformal Prediction Beyond Exchangeability

- Barber, Candès, Ramdas & Tibshirani (2023): coverage without exchangeability
  - the coverage gap is bounded by weighted total-variation distances
  - between the score vector and the same vector with the test score swapped with calibration score i
- How large these distances are under volatility clustering is an empirical question
- Gibbs, Cherian & Candès (2025): exact coverage conditional on a pre-specified class of covariate shifts is attainable; which class captures a financial crisis is open
- Choosing the ACI step γ: Zaffran et al. (2022) aggregate ACI experts over a grid of γ values

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Case Study, Further Key Results

Finding	Source	Reading
Strict and Auxiliary ESR sizes differ by at most 0.01 for every process and sample size, including AR(1)-GARCH with $\phi = 0.5$	Table 1	In these designs, dropping the VaR forecasts barely changes the size; power is compared separately (Fig. 2–3)
Student-t, 250-day window: the 0.12 share comes from GARCH with a yearly refit; from 500 days GJR-GARCH-t and GARCH-t differ by at most 0.01	Table 3	Leverage in the variance equation barely changes the rejection shares once the window is long

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Case Study, Limitations and Extensions

- Limits stated by the authors
  - Auxiliary and Strict ESR are two-sided only (paper, Sec. 1.3)
  - The classical covariance over-rejects in small samples
- Replication with the course data (Seminar 8, C3)
  - Run the Strict ESR (two-sided) and the Intercept ESR (one-sided) on the ES 2.5% forecasts of the six models in the four markets; compare with Z_2 and McNeil–Frey
  - Extend the Table 3 replication: GJR-GARCH, more stocks and the period after August 2019
- Open: a one-year supervisory window
  - At $T = 250$ the two-sided Intercept ESR has size 0.04 with the robust covariance and 0.15 with the classical one (Table 1, EGARCH-t); the one-sided size is reported only for $T = 2\,500$ (0.02, Table 2)
  - What is the one-sided size at one year?
  - Can a bootstrap of the t-statistic hold it at 5% without losing the power of Fig. 4?

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Scorecard Across Four Markets

- Each cell: breach rate of VaR 1% in % / Z_2 p-value / MCS p-value (FZ0)

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	1.67 / <0.001 / <0.001	1.17 / 0.134 / <0.001	0.77 / 0.863 / n/a	0.51 / 1.000 / n/a
Normal	2.70 / <0.001 / <0.001	1.90 / <0.001 / <0.001	1.74 / 0.001 / n/a	1.06 / 0.973 / n/a
Student-t	2.00 / 0.002 / <0.001	1.58 / 0.003 / <0.001	0.71 / 0.625 / n/a	0.85 / 0.920 / n/a
GARCH-t	1.77 / <0.001 / 0.010	1.46 / <0.001 / 0.096	n/a / n/a / n/a	n/a / n/a / n/a
FHS	1.23 / 0.209 / 1.000	1.20 / 0.236 / 1.000	n/a / n/a / n/a	n/a / n/a / n/a
GARCH-EVT	1.15 / 0.145 / 0.211	0.97 / 0.215 / 0.966	n/a / n/a / n/a	n/a / n/a / n/a

- n/a: withheld GARCH forecast; on Bitcoin and EUR/RON the MCS is not computed, because it needs all six models (slide on withheld forecasts)

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Answers to the Numeric Self-Checks

1. $\hat{\pi} = 0.006$; $\ln L(\alpha) = 497 \ln 0.99 + 3 \ln 0.01 = -18.81$, $\ln L(\hat{\pi}) = 497 \ln 0.994 + 3 \ln 0.006 = -18.34$
   - $\mathrm{LR}_{uc} = 2(-18.34 - (-18.81)) = 0.94 < 3.84$, $p = 0.33$: not rejected; 5 breaches were expected
2. $L = 3\%$: $\frac{\mathbf{3-2}}{\mathbf{0.05 \cdot 2.5}} + \frac{\mathbf{2}}{\mathbf{2.5}} + \ln 2.5 - 1 = 8 + 0.8 + 0.916 - 1 = 8.716$; $L = 1\%$: $0.8 + 0.916 - 1 = 0.716$
   - in decimals only $\ln e$ changes, by $\ln 0.01 = -4.605$: 4.111 and -3.889; the difference between the two days is the same
3. $\lambda = 1000 \cdot 0.01^{\mathbf{2}}/(0.01 \cdot 0.99) = 10.10$, $\sqrt{\lambda} = 3.178$
   - power $= \Phi(1.218) + \Phi(-5.138) = 0.888$ (the second term is below $10^{-\mathbf{6}}$): the 88.8% of the dashed $T = 1000$ curve; exact power 78%
4. $\sum_t L_t I_t / \mathrm{ES}_t = 5.5$, $N_T = 5$, $T\alpha = 6.25$
   - $Z_{\mathbf{1}} = 1 - 5.5/5 = -0.10$: on breach days losses are on average 10% deeper than the ES forecast
   - $Z_{\mathbf{2}} = 1 - 5.5/6.25 = 0.12 > 0$: fewer breaches than the 6.25 expected offset that depth; far from the 5% threshold -0.70

Self-checks

The Discovery Loop (1/2): Prompts

- Prompts (tools: Semantic Scholar, Elicit, Google Scholar; check every DOI on Crossref)
  - `List peer-reviewed work since 2019 on conformal prediction under serial dependence, with DOI, exact guarantee and assumptions.`
  - `Give three rival explanations for crisis over-breaches of split conformal VaR and what each predicts for breach rates by regime.`
  - `Act as a hostile referee: list the strongest reasons this crisis effect is an artefact and the test that would settle each.`

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The Discovery Loop (2/2): Checks the Researcher Runs

- **Look-ahead bias**
  - change the observations after day $t-1$ and confirm that VaR_t, the scale and the crisis flag do not move
- **Dependent breaches**
  - Christoffersen and duration tests, or a block bootstrap of the hit sequence
- **Power**
  - 613 crisis days give 6.1 expected breaches of VaR 1%
  - a true 2% rate is detected with probability 57% by an exact two-sided Binomial test at 5% (i.i.d. breaches assumed)
- **Multiplicity**: 4 markets $\times$ 2 levels $\times$ 3 methods $=$ 24 tests
  - Holm controls the FWER (family-wise error rate)
  - Benjamini–Hochberg controls the FDR (false discovery rate)
- **Misstated theorems**
  - e.g. "ACI guarantees conditional coverage"
- **Off-by-one quantiles**
  - $k = \lceil (n+1)(1-\alpha) \rceil$ is not the default of `np.quantile`

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Project Seed, Crisis-Conditional Coverage

- Questions
  - can a distribution-free VaR 1% and ES 2.5% forecast achieve crisis-conditional coverage?
  - does this coverage hold in all four markets under a pre-registered test?
- Replicate first: split conformal VaR 5% crisis rates (S&P 500 7.29%, EUR/RON 11.43%) and the ACI share of infinite VaR 1% (13.4% of all S&P 500 evaluation days)
- Deliverables
  - A pre-registration (hypotheses, tests, corrections, power) dated before the first run
  - A candidate method, e.g. Gibbs, Cherian & Candès (2025) with the volatility regime as conditioning class; VaR tests by regime; ES tested with the ES regression backtest (Bayer & Dimitriadis (2022)) or Z_2
  - Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

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Acronyms used in this material (1/2)

- **ACI**: Adaptive Conformal Inference
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model), as in AR(1)-GARCH
- **BCBS**: Basel Committee on Banking Supervision
- **BET**: Bucharest Exchange Trading (index)
- **BIS**: Bank for International Settlements
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **COVID-19**: COronaVIrus Disease 2019
- **DGP**: Data-Generating Process
- **DM**: Diebold–Mariano (test)
- **DQ**: Dynamic Quantile (test)
- **EGARCH**: Exponential GARCH
- **ES**: Expected Shortfall
- **ESR**: Expected Shortfall Regression (backtest)
- **EVT**: Extreme Value Theory
- **FDR**: False Discovery Rate
- **FHS**: Filtered Historical Simulation
- **FRTB**: Fundamental Review of the Trading Book

Acronyms used in this material (2/2)

- **FWER**: Family-Wise Error Rate
- **FZ**: Fissler–Ziegel (family of scoring functions for VaR and ES)
- **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-EVT**: GARCH filter combined with Extreme Value Theory for the tail
- **GAS**: Generalised Autoregressive Score (model)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **GPD**: Generalised Pareto Distribution
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HS**: Historical Simulation
- **IMA**: Internal Models Approach
- **LLM**: Large Language Model
- **MCS**: Model Confidence Set
- **OLS**: Ordinary Least Squares
- **PIT**: Probability Integral Transform
- **POF**: Proportion Of Failures (test)
- **QLIKE**: Quasi-LIKElihood (loss function)

References (1/4)

- Acerbi, C., Szekely, B. (2014). Backtesting expected shortfall. MSCI Inc., December 2014.
- Angelopoulos, A.N., Bates, S. (2021). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511.
- Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J. (2023). Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2).
- Basel Committee on Banking Supervision (1996). Supervisory framework for the use of "backtesting" in conjunction with the internal models approach to market risk capital requirements. Bank for International Settlements.
- Basel Committee on Banking Supervision (2019). Minimum capital requirements for market risk (rev. February 2019). Bank for International Settlements.
- Bayer, S., Dimitriadis, T. (2022). Regression-based expected shortfall backtesting. *Journal of Financial Econometrics*, 20(3), 437–471.
- Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- Christoffersen, P., Pelletier, D. (2004). Backtesting value-at-risk: a duration-based approach. *Journal of Financial Econometrics*, 2(1), 84–108.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.

References (2/4)

- Du, Z., Escanciano, J.C. (2017). Backtesting expected shortfall: accounting for tail risk. *Management Science*, 63(4), 940–958.
- Ehm, W., Gneiting, T., Jordan, A., Krüger, F. (2016). Of quantiles and expectiles: consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society, Series B*, 78(3), 505–562.
- Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- Escanciano, J.C., Olmo, J. (2010). Backtesting parametric value-at-risk with estimation risk. *Journal of Business & Economic Statistics*, 28(1), 36–51.
- Fissler, T., Ziegel, J.F. (2016). Higher order elicitability and Osband's principle. *Annals of Statistics*, 44(4).
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- Gibbs, I., Candès, E. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, 34; arXiv:2106.00170.
- Gibbs, I., Cherian, J.J., Candès, E.J. (2025). Conformal prediction with conditional guarantees. *Journal of the Royal Statistical Society, Series B*, 87(4), 1100–1126.

References (3/4)

- Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (eds.) (2017). Applied Quantitative Finance, 3rd ed. Springer.
- Hoga, Y., Demetrescu, M. (2023). Monitoring value-at-risk and expected shortfall forecasts. *Management Science*, 69(5), 2954–2971.
- Kratz, M., Lok, Y.H., McNeil, A.J. (2018). Multinomial VaR backtests: a simple implicit approach to backtesting expected shortfall. *Journal of Banking & Finance*, 88, 393–407.
- Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R.J., Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523), 1094–1111.
- McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

References (4/4)

- Nolde, N., Ziegel, J.F. (2017). Elicitability and backtesting: perspectives for banking regulation. *Annals of Applied Statistics*, 11(4).
- Patton, A.J., Ziegel, J.F., Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2), 388–413.
- Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- Pele, D.T., Lazar, E., Mazurencu-Marinescu-Pele, M. (2019). Modeling expected shortfall using tail entropy. *Entropy*, 21(12), 1204.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Finite-sample precision limits for expected shortfall forecast comparisons. *Mathematics*, 14(13), 2316.
- Zaffran, M., Féron, O., Goude, Y., Josse, J., Dieuleveut, A. (2022). Adaptive conformal predictions for time series. *Proceedings of the 39th International Conference on Machine Learning*; arXiv:2202.07282.